

ASCEND INSTITUTE, GANGTOK
CHAPTER-WISE TEST
ICSE CLASS 8 - CHEMISTRY

Chapter: Language of Chemistry

Date: _____

Name: _____

Maximum Marks: 20

Time: 40 minutes

Instructions: Answer all questions. Section A contains 8 MCQs of 1 mark each. Section B contains 6 questions of 2 marks each. Show the necessary steps wherever required.

SECTION A - 8 MARKS

1. The correct formula formed by Na^+ and O^{2-} is:

- (i) NaO
(k) NaO_2

- (j) Na_2O
(l) Na_2O_2

Answer: _____

2. The correct formula of aluminium sulphate is:

- (i) AlSO_4
(k) $\text{Al}_2(\text{SO}_4)_3$

- (j) Al_2SO_4
(l) $\text{Al}_3(\text{SO}_4)_2$

Answer: _____

3. Calcium has valency 2 and hydroxide has valency 1. The formula of calcium hydroxide is:

- (i) CaOH
(k) $\text{Ca}(\text{OH})_2$

- (j) Ca_2OH
(l) $\text{Ca}_2(\text{OH})_2$

Answer: _____

4. How many oxygen atoms are present in $\text{Ca}(\text{NO}_3)_2$?

- (i) 2
(k) 5

- (j) 3
(l) 6

Answer: _____

5. The formula Al_2O_3 is consistent with valencies:

- (i) $\text{Al} = 2, \text{O} = 3$
(k) $\text{Al} = 1, \text{O} = 3$

- (j) $\text{Al} = 3, \text{O} = 2$
(l) $\text{Al} = 3, \text{O} = 1$

Answer: _____

6. While balancing an equation, one should change:

- (i) Only subscripts
(k) The formula of each compound

- (j) Only coefficients
(l) Names of substances

Answer: _____

7. Which equation is balanced?

- (i) $\text{H}_2 + \text{O}_2 \rightarrow \text{H}_2\text{O}$
(k) $\text{H}_2 + 2\text{O}_2 \rightarrow \text{H}_2\text{O}$

- (j) $2\text{H}_2 + \text{O}_2 \rightarrow 2\text{H}_2\text{O}$
(l) $2\text{H}_2 + 2\text{O}_2 \rightarrow \text{H}_2\text{O}$

Answer: _____

8. The correct formula of ammonium sulphate is:

- (i) NH_4SO_4
(k) $\text{NH}_4(\text{SO}_4)_2$

- (j) $(\text{NH}_4)_2\text{SO}_4$
(l) $(\text{NH}_4)_2(\text{SO}_4)_2$

Answer: _____

SECTION B - 12 MARKS

Answer all questions. Continue on the back side if additional space is required.

9. Write the formula of aluminium sulphate using the valencies Al = 3 and $\text{SO}_4 = 2$. [2]

10. Write the formula of magnesium hydroxide using Mg = 2 and OH = 1. [2]

11. In Fe_2O_3 , oxygen has valency 2. Find the valency of iron. [2]

12. Find the total number of atoms of each element in $\text{Al}_2(\text{SO}_4)_3$. [2]

13. Balance: $\text{H}_2 + \text{O}_2 \rightarrow \text{H}_2\text{O}$. [2]

14. A student writes magnesium chloride as MgCl. Correct the formula and explain why. [2]
